July 2025 CSE 208

Online on APSP

Time: 30 minutes

A logistics company manages a delivery network represented as a weighted directed graph. Each node is a city, and each edge is a direct road with a specific cost (fuel/tolls). To ensure quality control, the company has mandated that every delivery route must pass through one of their two regional distribution centers, located at **city Y and Z**, regardless of whether it is the most efficient path.

Calculate the shortest possible cost to travel between all pairs of cities (i,j) under the strict constraint that the **path must include either city Y or city Z**.

Input

The first input line has two integers n and m: the number of cities and roads ($1 <= n <= 500$, $1 <= m <= n^2$).

Then, there are m lines describing the roads. Each line has three integers a, b and c: there is a road between cities a and b whose length is c ($1 <= a,b <= n$, $1 <= c <= 10^9$).

Next line will take the city Y and Z

Next line contains the number of queries q

Finally, there are q lines describing the queries. Each line has two integers a and b: determine the length of the shortest route between cities a and b.

Output

Print the length of the shortest route for each query. If there is no route, print -1 instead.

Example

Sample Input	Sample Output
4 5 0 2 5 0 1 10 1 2 2 0 3 5 3 2 2 1 3 2 0 2 1 2	7 2

5 7 0 1 2 0 2 10 1 2 3 2 3 1 1 3 10 0 3 2 3 4 1 1 3 3 0 4 1 3 0 2	3 3 4